

Mathematical Statistics (I)

Final Exam.

June 13, 2016

1.(80) Let X and Y have joint pmf

$$p(x, y) = \frac{e^{-2}}{x!(y-x)!}, \quad x = 0, 1, \dots, y; y = 0, 1, 2, \dots$$

(1)(40) Find the joint mgf $M(t_1, t_2)$.

(2)(40) Compute $E(X|y)$.

2.(60) Assume that $X \sim F(r, r)$. Derive the pdf of $Y = 1/X$.

3.(40) Let $X_1, X_2, \dots, X_n$ be r.s. from $P(\lambda)$. Find a function $g(\cdot)$ s.t. $\sqrt{n}(g(\bar{X}) - g(\lambda))$ converges in distribution to the standard normal distribution.

4.(120) Let $X_1, X_2, \dots, X_n$ be r.s. from $U(0, \theta)$, and let $Y_n = \max\{X_1, \dots, X_n\}$.

(1)(40) Find the limiting distribution of Y_n .

(2)(40) Using the result of (1), show that Y_n is a consistent estimator of θ .

(3)(40) Using the WLLN, find a consistent estimator of θ based on $X_1, X_2, \dots, X_n$.

5.(140) Let $X_1, X_2, \dots, X_n$ be r.s. from $N(\mu, \sigma^2)$. Let $\bar{X} = \sum X_i/n$ and $s^2 = \sum (X_i - \bar{X})^2/(n-1)$.

(1)(40) Find the exact distribution of $n(\bar{X} - \mu)^2/s^2$.

(2)(40) Show that the limiting distribution of $\sqrt{n}(\bar{X} - \mu)/s$ is $N(0, 1)$.

(3)(60) Compute $E[\bar{X}^2/s^2]$.

6.(60) Let $\{\mathbf{x}_n\}$ be a sequence of iid random vectors with $E(\mathbf{x}_1) = \mathbf{0}$ and $Cov(\mathbf{x}_1) = \Sigma$. Show that $\mathbf{y}_n = \sqrt{n}\bar{\mathbf{x}}$ converges in distribution to $N_p(\mathbf{0}, \Sigma)$.